space. This is a source of O_2, certainly, but it doesn't easily account for so *much* oxygen.

Another possibility is that ordinary visible light, which the Sun pours out in vast amounts, is used on Earth to break water apart—except that there's no known way to do this without life. There would have to be plants—life-forms colored by a pigment that strongly absorbs visible light, that knows how to split a water molecule by saving up the energy of two photons of light, that retains the H and excretes the O, and that uses the hydrogen thus liberated to synthesize organic molecules. The plants would have to be spread over much of the planet. All this is asking a lot. If you're a good skeptical scientist, so much O_2 would not be proof of life. But it certainly might be cause for suspicion.

With all that oxygen you're not surprised to discover ozone (O_3) in the atmosphere, because ultraviolet light makes ozone out of molecular oxygen (O_2). The ozone then absorbs dangerous ultraviolet radiation. So if the oxygen *is* due to life, there's a curious sense in which the life is protecting itself. But this life might be mere photosynthetic plants. A high level of intelligence is not implied.

When you examine the continents more closely, you find there are, crudely speaking, two kinds of regions. One shows the spectrum of ordinary rocks and minerals as found on many worlds. The other reveals something unusual: a material, covering vast areas, that strongly absorbs red light. (The Sun, of course, shines in light of all colors, with a peak in the yellow.) This pigment might be just the agent needed if ordinary visible light *is* being used to break water apart and account for the oxygen in the air. It's another hint, this time a little stronger, of life—not a bug here and there, but a planetary surface overflowing with life. The pigment is in fact chlorophyll: It absorbs blue light as well as red, and is responsible for the fact that plants are green. What you're seeing is a densely vegetated planet.

So the Earth is revealed to possess three properties unique at least in this solar system—oceans, oxygen, life. It's hard not to think